Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania

1. Condensed original-idea version

1.1 Research Motivation

Standard competitive labor-market theory predicts that a higher minimum wage will cause employers to cut low-wage employment (Stigler, 1946). However, empirical studies reach mixed conclusions. Older aggregate studies using youth employment data tended to support this negative prediction. By contrast, newer comparative case studies found little or no job loss after minimum-wage rises, including research on California, the federal U.S. minimum-wage change, and minimum-wage policies in Britain.

Many previous analyses rely on aggregate data, which struggle to separate policy effects from macroeconomic disturbances. Hence, more reliable firm-level evidence from a clear quasi-natural experiment is needed.

New Jersey's 1992 minimum-wage hike created such an opportunity. On April 1, 1992, New Jersey raised its state minimum wage from $4.25 to $5.05 per hour, while neighboring eastern Pennsylvania kept its minimum wage unchanged. Importantly, this policy was implemented during a recession, so economic boom would not cover up the true employment impact of minimum-wage policy. Fast-food restaurants, which hire large numbers of minimum-wage workers, are chosen as the research subject.

1.2 Research Objective

This paper estimates the causal impacts of a binding minimum-wage increase at the establishment level. It focuses on key outcomes including employment, wages, consumer prices and non-wage benefits.

Two identification strategies are adopted. First, difference-in-differences compares fast-food stores in New Jersey (treatment group) with stores in eastern Pennsylvania (control group). Second, the study conducts within-state comparisons inside New Jersey: it uses a wage-gap (GAP) measure to compare stores that faced different magnitudes of required wage increases. The authors also test whether firms offset higher wage costs by cutting fringe benefits or adjusting wage growth, and check whether higher labor costs are passed to consumers via product prices.

Core research question: Does a binding minimum-wage increase reduce employment for low-wage fast-food businesses?

2. Data Source, Sample Construction and Variable Definitions

2.1 Data Source

This study uses original two-wave survey data for fast-food restaurants.

Wave 1 (pre-policy): Feb-Mar 1992, before New Jersey's minimum-wage increase.

Wave 2 (post-policy): Nov-Dec 1992, about eight months after the policy change.

Surveyed chains: Burger King, KFC, Roy Rogers, Wendy's. McDonald's was excluded for low response rates.

Treatment group: restaurants in New Jersey.

Control group: restaurants in eastern Pennsylvania, where the minimum wage remained unchanged.

Researchers conducted on-site visits for non-respondents to confirm store closure status.

2.2 Sample Construction

The initial sampling frame included 473 restaurants. 410 stores completed Wave-1 interviews (response rate 86.7%).

All 410 stores were tracked in Wave 2: 6 permanently closed, 2 temporarily closed, 2 under renovation. Permanently-closed stores: post-policy FTE employment set to 0. Temporarily-closed / renovated stores: employment set to missing in baseline regressions (treated as closed only in robustness tests).

Baseline regression sample (Table 4): 357 stores, requiring non-missing employment and starting-wage data in both waves.

Price regression sample (Table 7): 315 stores, further restricted to valid meal-price observations.

2.3 Variable Definitions

2.3.1Dependent Variables

（1） fte1, fte2: Full-time-equivalent (FTE) employment before and after the policy.
The calculation formula is: FTE = managers + full-time non-manager workers + 0.5 part-time workers

（2） demp: Change in FTE employment, formula is demp = fte2 - fte1. This is the main outcome variable for employment regressions.

（3） dln_pmeal: Log change of the "full-meal" price (medium soda, small fries, main dish), formula is dln_pmeal=ln(pmeal2)-ln(pmeal). Used for price-effect estimation.

2.3.2 Key Independent Variables

(1) nj: Dummy variable, 1 = store in New Jersey (treatment), 0 = store in Pennsylvania (control), for difference-in-differences estimation.

(2) gap: Continuous measure of policy exposure.

$$gap_i = \begin{cases} 0 \\ \dfrac{5.05 - w_{1i}}{w_{1i}} \end{cases}$$

Where PA stores, and PJ stores with pre-policy wage >=5.05, gap = 0. It is the proportional wage increase needed to reach the new minimum wage. Larger values mean stronger policy impact.

2.3.3 Control Variables

(1) Chain dummies (bk, kfc, roys): restaurant brand identifiers.

(2) co_owned: Dummy, 1 = company-owned store, 0 = franchise store.

(3) Regional dummies: absorb local economic shocks.

3. Empirical Design

This study exploits a quasi-natural experiment: New Jersey raised its state minimum wage from $4.25 to $5.05 per hour on April 1, 1992, while the minimum wage in neighboring eastern Pennsylvania remained unchanged at $4.25. Two complementary identification strategies are applied to estimate the causal effect of minimum-wage increases on employment.

3.1 Identification Strategy 1: Difference-in-Differences (DID)

The first strategy compares changes in employment at fast-food restaurants in New Jersey (treatment group, nj=1) with restaurants in eastern Pennsylvania (control group, nj=0).

The baseline DID regression specification is:

$$\Delta E_i = \alpha + cnj_i + X_i\beta + \varepsilon_i$$

Where E (demp) is the change in full-time-equivalent (FTE) employment for store i; nj is dummy variable equal to 1 for New Jersey stores; X is vector of control variables, including chain dummies, company-owned store dummy, and regional dummies; Coefficient c captures the relative change in employment caused by the minimum-wage hike.

The key identifying assumption is the parallel-trends assumption: absent the minimum-wage policy, employment growth would have been similar for restaurants in New Jersey and Pennsylvania.

3.2 Identification Strategy 2: Within-state continuous treatment (Wage-gap / GAP estimator)

To avoid reliance on cross-state comparisons alone, the second identification uses within-New-Jersey variation in how much each store needed to raise wages to meet the new legal minimum. A variable gap measures the proportional wage increase required at each store. Larger gap means the store faces stronger policy pressure to lift wages.

3.3 Robustness Checks and Placebo Test

3.3.1 Robustness Checks

A series of robustness tests are implemented to assess result sensitivity: re-coding temporary store closures, modifying the FTE employment formula by adjusting part-time worker weights or excluding managers, restricting the sample to reduce survey response bias, and weighted regression using pre-policy employment size.

3.3.2 Placebo Test

A placebo falsification test is further conducted. We limit the sample to Pennsylvania stores only and generate an artificial fake-gap measure. If the estimation framework suffers from inherent spurious correlation, this constructed variable would predict employment shifts. Valid identification requires the placebo coefficient to be near zero and statistically insignificant. Beyond employment, reduced-form price regressions estimate whether higher labor costs from the minimum-wage reform were passed on to consumers via meal prices.

4. Core replication results and comparation with the original paper

4.1 Group-wise descriptive statistics

-> nj = NJ

Variable	Obs	Mean	Std. dev.	Min	Max
fte1	321	20.43941	9.106239	5	85
fte2	320	20.96172	9.352609	0	60.5
demp	310	.4070968	8.503438	-34	34
wage_st	314	4.612134	.3463505	4.25	5.75
wage_st2	318	5.080849	.1045445	5	5.75
gap	331	.0995381	.0734713	0	.1882353

-> nj = PA

Variable	Obs	Mean	Std. dev.	Min	Max
fte1	77	23.33117	11.85628	7.5	70.5
fte2	77	21.16558	8.276732	0	43.5
demp	75	-2.283333	10.85363	-41.5	22.75
wage_st	76	4.630132	.3516873	4.25	5.5
wage_st2	71	4.617465	.3574781	4.25	6.25
gap	79	0	0	0	0

Two-sample t test with equal variances

Group	Obs	Mean	Std. err.	Std. dev.	[95% conf.	interval]
PA	77	23.33117	1.351149	11.85628	20.64012	26.02222
NJ	321	20.43941	.5082607	9.106239	19.43945	21.43936
Combined	398	20.99887	.4887136	9.749805	20.03808	21.95966
diff		2.891761	1.230207		.4732069	5.310315

diff = mean(PA) - mean(NJ) t = 2.3506
H0: diff = 0 Degrees of freedom = 396

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.9904 Pr(|T| > |t|) = 0.0192 Pr(T > t) = 0.0096

This is a part of my replication about the description statistics, compared with the original paper.

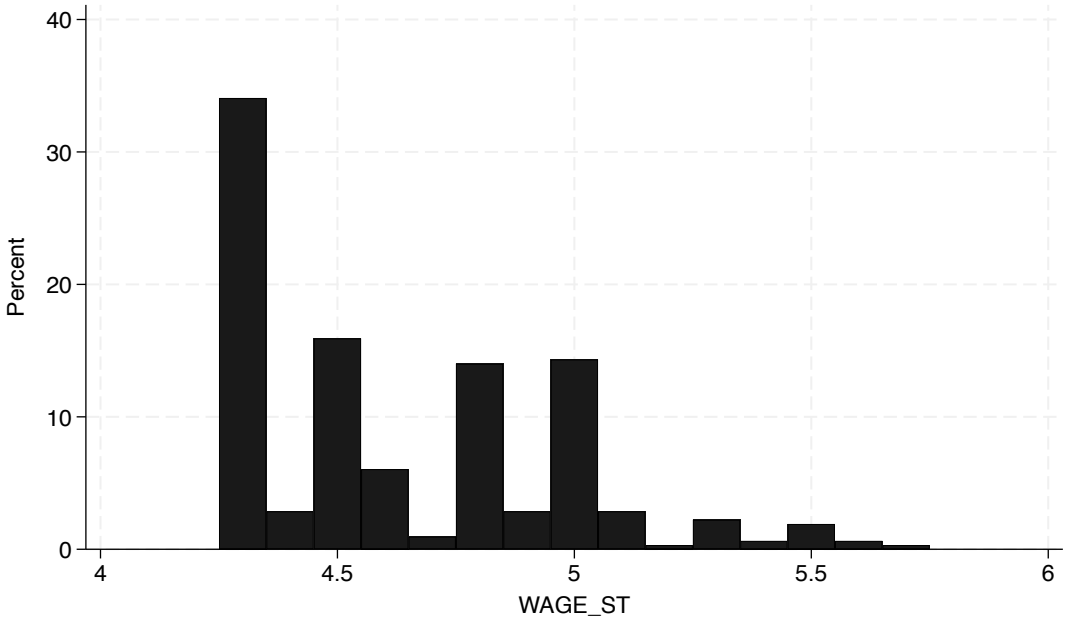
TABLE 1—SAMPLE DESIGN AND RESPONSE RATES

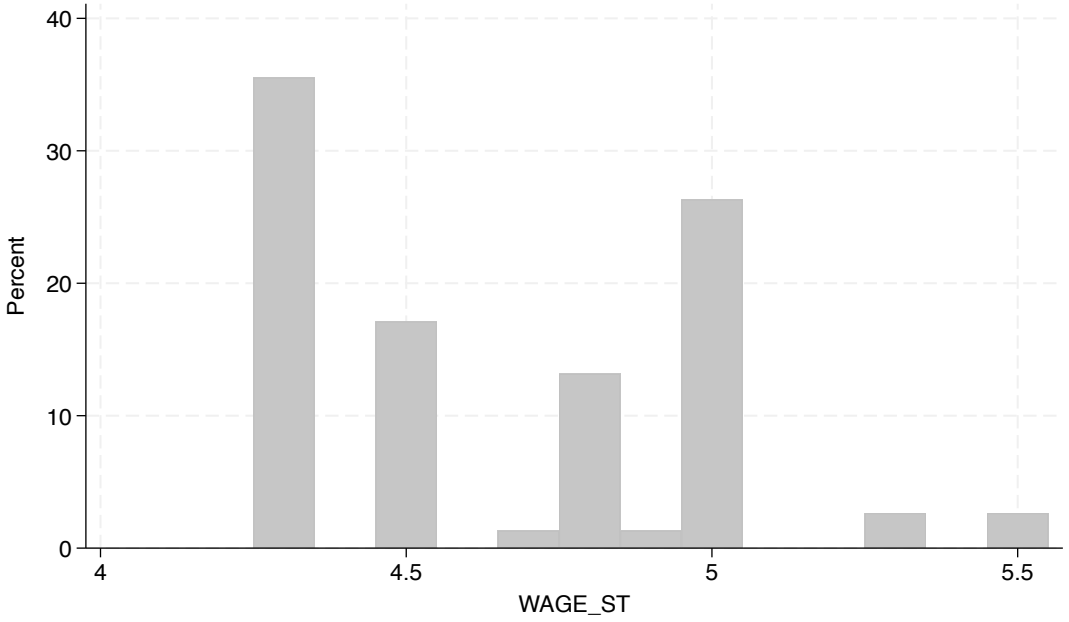
	All	Stores in: NJ	Stores in: PA
Wave 1, February 15–March 4, 1992:			
Number of stores in sample frame:[a]	473	364	109
Number of refusals:	63	33	30
Number interviewed:	410	331	79
Response rate (percentage):	86.7	90.9	72.5
Wave 2, November 5–December 31, 1992:			
Number of stores in sample frame:	410	331	79
Number closed:	6	5	1
Number under rennovation:	2	2	0
Number temporarily closed:[b]	2	2	0
Number of refusals:	1	1	0
Number interviewed:[c]	399	321	78

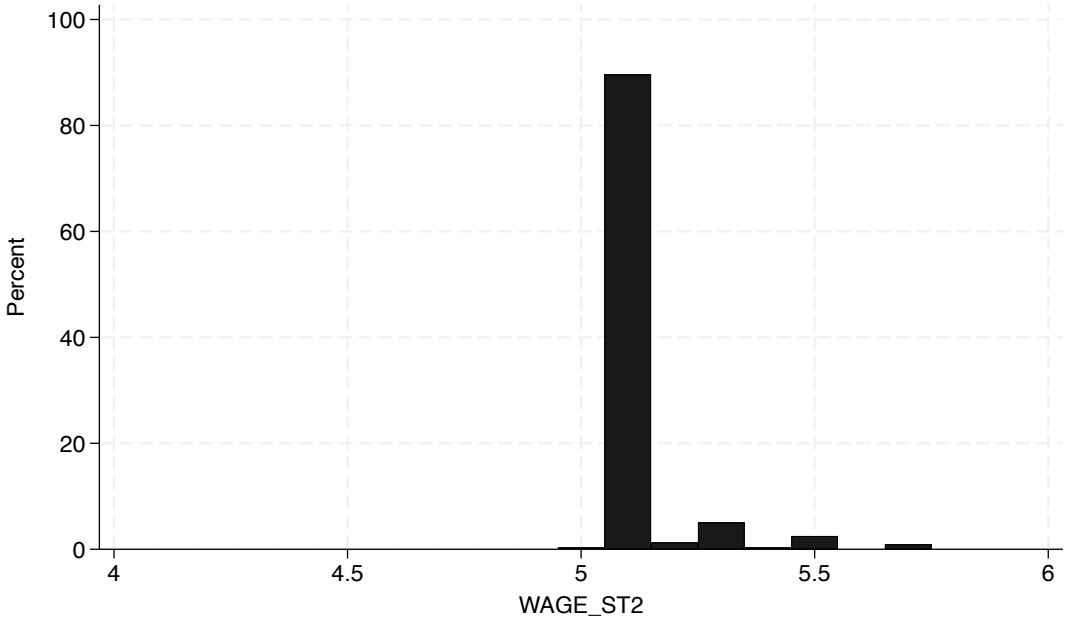
TABLE 2—MEANS OF KEY VARIABLES

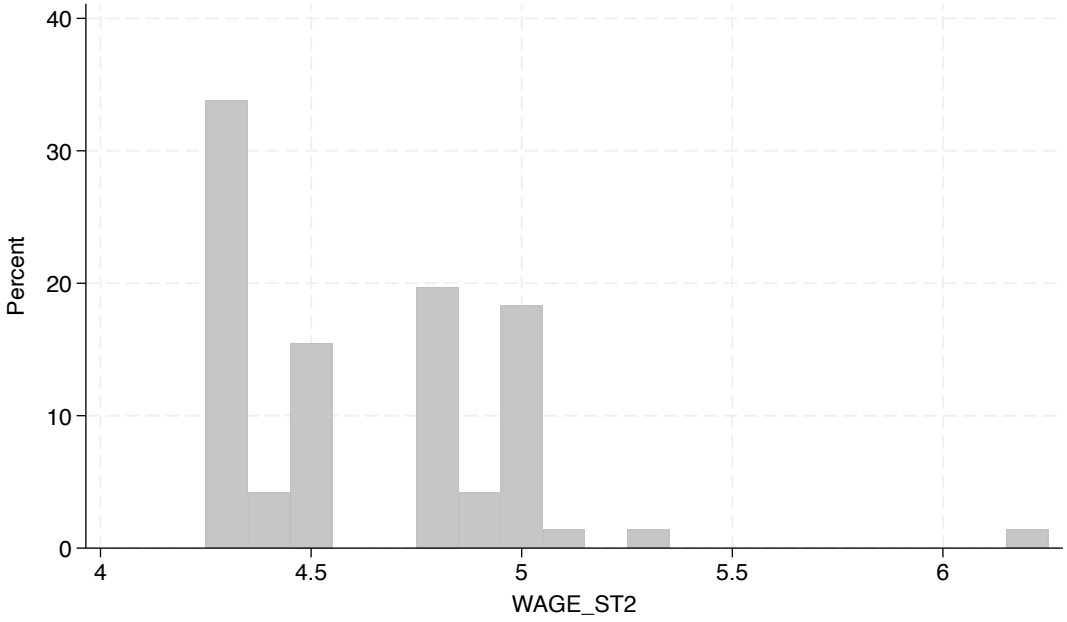
Variable	Stores in: NJ	PA	t[a]
1. *Distribution of Store Types (percentages):*			
a. Burger King	41.1	44.3	−0.5
b. KFC	20.5	15.2	1.2
c. Roy Rogers	24.8	21.5	0.6
d. Wendy's	13.6	19.0	−1.1
e. Company-owned	34.1	35.4	−0.2
2. *Means in Wave 1:*			
a. FTE employment	20.4	23.3	−2.0
	(0.51)	(1.35)	
b. Percentage full-time employees	32.8	35.0	−0.7
	(1.3)	(2.7)	
c. Starting wage	4.61	4.63	−0.4
	(0.02)	(0.04)	
d. Wage = $4.25 (percentage)	30.5	32.9	−0.4
	(2.5)	(5.3)	
e. Price of full meal	3.35	3.04	4.0
	(0.04)	(0.07)	
f. Hours open (weekday)	14.4	14.5	−0.3
	(0.2)	(0.3)	
g. Recruiting bonus	23.6	29.1	−1.0
	(2.3)	(5.1)	
3. *Means in Wave 2:*			
a. FTE employment	21.0	21.2	−0.2
	(0.52)	(0.94)	
b. Percentage full-time employees	35.9	30.4	1.8
	(1.4)	(2.8)	
c. Starting wage	5.08	4.62	10.8
	(0.01)	(0.04)	
d. Wage = $4.25 (percentage)	0.0	25.3	—
		(4.9)	
e. Wage = $5.05 (percentage)	85.2	1.3	36.1
	(2.0)	(1.3)	
f. Price of full meal	3.41	3.03	5.0
	(0.04)	(0.07)	
g. Hours open (weekday)	14.4	14.7	−0.8
	(0.2)	(0.3)	
h. Recruiting bonus	20.3	23.4	−0.6
	(2.3)	(4.9)	

The histogram replication is:

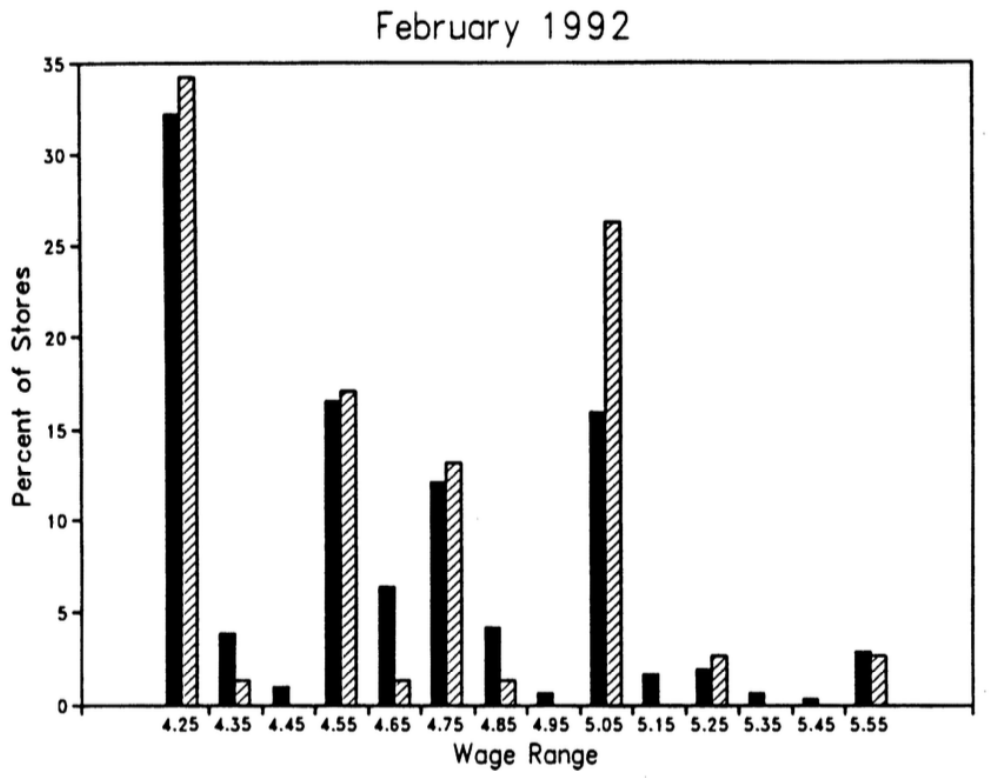

Percent
0
10
20
30
40
4
4.5
5
5.5
6
WAGE_ST

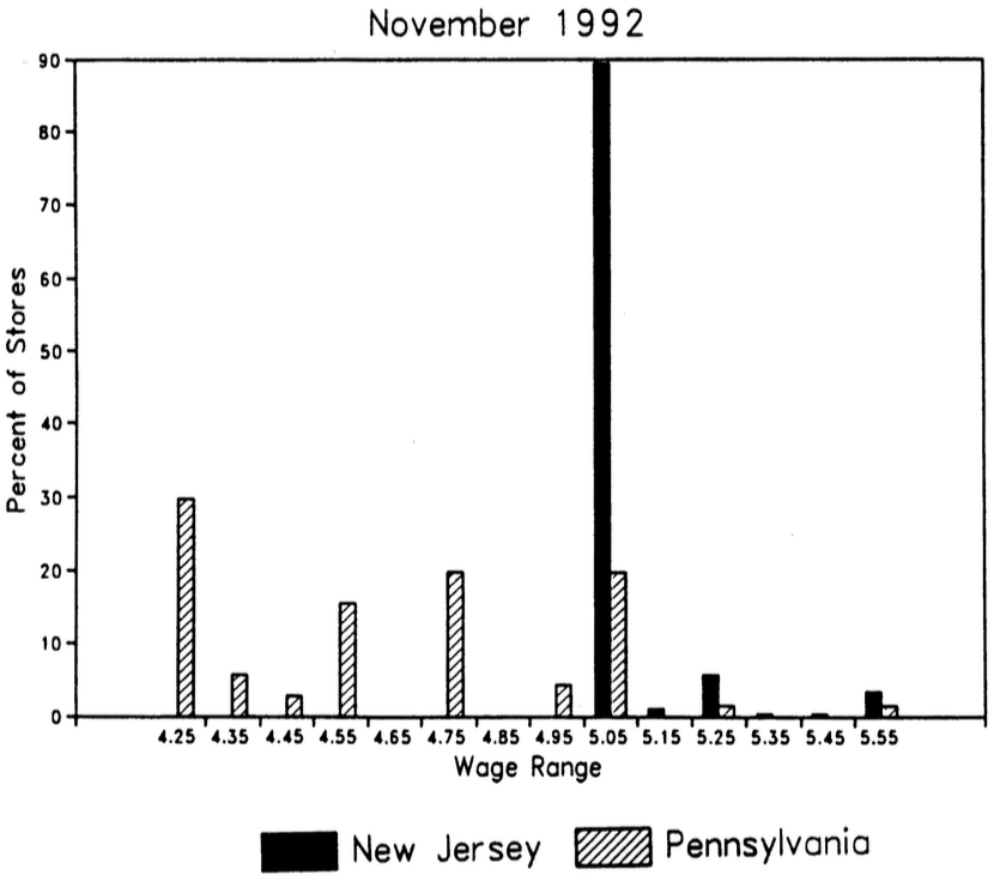

Percent
0
10
20
30
40
4
4.5
5
5.5
WAGE_ST





They are almost the same with the original paper in February and November 1992:





4.2 Differences in Differences

wage_group	fte1	fte2	demp
0	77	77	75
	23.33117	21.16558	-2.283333
	11.85628	8.276732	10.85363
1	97	98	94
	19.5567	20.87755	1.204787
	7.631444	9.957917	7.974334
2	137	135	132
	20.08066	20.95556	.7098485
	9.846855	8.779368	7.884145
3	87	87	84
	21.98851	21.06609	-.9613095
	9.311151	9.617576	9.852718
Total	398	397	385
	20.99887	21.00126	-.117013
	9.749805	9.144222	9.056784

Simple DID = Average employment change of NJ stores − Average employment change of PA stores:

$$\Delta_{NJ} = \frac{97 \times 1.204787 + 137 \times 0.7098485 + 87 \times (-0.9613095)}{321} \approx 0.4065$$

$$DID = \Delta_{NJ} - \Delta_{PA} = 0.4065 - (-2.2833) = 2.6898 \approx 2.69$$

The original DID is 2.76

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE IN NEW JERSEY MINIMUM WAGE

Variable	Stores by state			Stores in New Jersey[a]			Differences within NJ[b]	
	PA (i)	NJ (ii)	Difference, NJ − PA (iii)	Wage = $4.25 (iv)	Wage = $4.26–$4.99 (v)	Wage ≥ $5.00 (vi)	Low–high (vii)	Midrange–high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	−2.69 (1.37)	−2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	−2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores[c]	−2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	−2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0[d]	−2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	−2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

This raw comparison shows that after the minimum wage hike, average employment in New Jersey restaurants rose relative to Pennsylvania restaurants. Within New Jersey, stores that faced the largest required wage increase (those paying exactly $4.25 before the reform) experienced employment gains, while stores already paying wages above the new minimum saw employment decline. This pattern contradicts the standard competitive labour market prediction that higher minimum wages reduce employment.

Variable	**PA (wage_group=0)**	**NJ low wage (wage_group=1) Wage=$4.25**	**NJ mid wage (wage_group=2) $4.26–$4.99**	**NJ high wage (wage_group=3) Wage≥$5.00**
Panel A: Replication Results (balanced sample, only stores with both waves)				
FTE employment before (fte1)	23.33	19.56	20.08	21.99
FTE employment after (fte2)	21.17	20.88	20.96	21.07
Change in mean FTE (demp)	-2.28	1.20	0.71	-0.96
N	77	97	137	87
Panel B: Original Card & Krueger Table3 (all available observations)				
FTE employment before	23.33	19.56	20.08	22.25
FTE employment after	21.17	20.88	20.96	20.21
Change in mean FTE	-2.16	1.32	0.87	-2.04
Simple DID estimate	**Replication: 2.69**	**Original: 2.76**	—	—

The replication estimate of 2.69 is very close to the original 2.76. The small gap comes from different sample definitions:

（1） The original paper uses all available observations: the pre-policy and post-policy means are computed separately, and do not require the same store to be surveyed in both waves. Some stores only appear in Wave 1 or Wave 2 and are still included in the corresponding mean.

（2） The balanced sample in this replication restricts the sample to stores with valid FTE values in both waves, dropping a small number of stores with missing values in either wave.

Qualitatively, the replication fully recovers the original finding. Pennsylvania stores experienced employment declines. New Jersey stores with the largest wage pressure had positive employment growth, while high-wage New Jersey stores had falling employment. The positive DID estimate remains robust: the minimum wage increase was associated with a relative rise in FTE employment in New Jersey fast-food restaurants.

4.3 Reduced-Form Models for Change in Employment

TABLE 4—REDUCED-FORM MODELS FOR CHANGE IN EMPLOYMENT

	Model				
Independent variable	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	2.33 (1.19)	2.30 (1.20)	—	—	—
2. Initial wage gap[a]	—	—	15.65 (6.08)	14.92 (6.21)	11.91 (7.39)
3. Controls for chain and ownership[b]	no	yes	no	yes	yes
4. Controls for region[c]	no	no	no	no	yes
5. Standard error of regression	8.79	8.78	8.76	8.76	8.75
6. Probability value for controls[d]	—	0.34	—	0.44	0.40

Table 4 presents reduced-form regressions of the change in FTE employment (demp) on the New Jersey treatment dummy (nj) or the initial wage gap (gap), with progressively added controls for restaurant chain, ownership type and regional fixed effects. All regressions use robust standard errors and the balanced sample of 351 stores. The dependent variable demp = fte2 − fte1, the change in full-time equivalent employment after the minimum wage reform.

Model (i) and (ii) use the binary DiD specification with a New Jersey dummy. Models (iii)-(v) use the continuous treatment specification with the initial wage gap, which measures how much the store's wage needed to rise to meet the new $5.05 minimum wage.

Model	Specification	My Replication Coefficient (Robust SE)	Original Card & Krueger Coefficient (SE)
(i)	reg demp nj, robust	2.277 (1.450)	2.33 (1.19)
(ii)	reg demp nj bk kfc roys co_owned, robust	2.282 (1.453)	2.30 (1.20)

Model	Specification	My Replication Coefficient (Robust SE)	Original Card & Krueger Coefficient (SE)
(iii)	reg demp gap, robust	17.051 (6.150)	15.65 (6.08)
(iv)	reg demp gap bk kfc roys co_owned, robust	16.363 (6.541)	14.92 (6.21)
(v)	reg demp gap bk kfc roys co_owned southj centralj northj pa2, robust	13.879 (7.062)	11.91 (7.39)

This replication broadly reproduces the core findings of Card and Krueger. The baseline binary difference-in-differences estimate of 2.277 closely tracks the original coefficient of 2.33, although the larger robust standard error yields weak statistical significance, consistent with the original study. Including chain and ownership controls barely shifts the treatment coefficient, and these added controls are statistically insignificant. For the continuous wage-gap specification, both replicated and original estimates are positive and significant: a one-dollar increase in the required wage gap raises FTE employment. After incorporating brand and ownership controls, the coefficient remains positive and significant at the 5% level with a modest reduction in magnitude. Introducing regional fixed effects further attenuates the treatment effect and weakens statistical significance. Across all five specifications, point estimates follow the same pattern reported in the original paper. Discrepancies are confined to minor differences in coefficients and standard errors, leaving the main qualitative conclusions unchanged.

Minor discrepancies between replication and original results arise primarily from standard error choices and subtle variable or sample coding differences. This replication applies robust standard errors, which inflate standard error magnitudes relative to the original specification and weaken statistical significance. Small differences in constructing the wage gap and FTE employment variables also shift point estimates slightly. As regional fixed effects absorb geographic heterogeneity, estimates become more sensitive to model specification. Nonetheless, the treatment effect maintains a positive sign and attenuates with additional controls, preserving the core economic conclusion of the original study.

4.4 Specification Tests

TABLE 5—SPECIFICATION TESTS OF REDUCED-FORM EMPLOYMENT MODELS

Specification	Change in employment		Proportional change in employment	
	NJ dummy (i)	Gap measure (ii)	NJ dummy (iii)	Gap measure (iv)
1. Base specification	2.30 (1.19)	14.92 (6.21)	0.05 (0.05)	0.34 (0.26)
2. Treat four temporarily closed stores as permanently closed[a]	2.20 (1.21)	14.42 (6.31)	0.04 (0.05)	0.34 (0.27)
3. Exclude managers in employment count[b]	2.34 (1.17)	14.69 (6.05)	0.05 (0.07)	0.28 (0.34)
4. Weight part-time as 0.4×full-time[c]	2.34 (1.20)	15.23 (6.23)	0.06 (0.06)	0.30 (0.33)
5. Weight part-time as 0.6×full-time[d]	2.27 (1.21)	14.60 (6.26)	0.04 (0.06)	0.17 (0.29)
6. Exclude stores in NJ shore area[e]	2.58 (1.19)	16.88 (6.36)	0.06 (0.05)	0.42 (0.27)
7. Add controls for wave-2 interview date[f]	2.27 (1.20)	15.79 (6.24)	0.05 (0.05)	0.40 (0.26)
8. Exclude stores called more than twice in wave 1[g]	2.41 (1.28)	14.08 (7.11)	0.05 (0.05)	0.31 (0.29)
9. Weight by initial employment[h]	—	—	0.13 (0.05)	0.81 (0.26)
10. Stores in towns around Newark[i]	—	33.75 (16.75)	—	0.90 (0.74)
11. Stores in towns around Camden[j]	—	10.91 (14.09)	—	0.21 (0.70)
12. Pennsylvania stores only[k]	—	−0.30 (22.00)	—	−0.33 (0.74)

Dependent variable: Change in FTE employment (demp), gap measure specification. Robust standard errors in parentheses for replication; conventional standard errors in parentheses for the original.

	Specification	**Replicated gap (robust SE)**	**Rep. N**	**Original gap (SE)**
1	Base specification (demp)	16.363 (6.541)	351	14.92 (6.21)
2	Treat temporarily closed as permanently closed (demp_r2)	16.363 (6.541)	351	14.42 (6.31)
3	Exclude managers in employment count (demp_nomgr)	15.786 (6.435)	351	14.69 (6.05)
4	Weight part-time as 0.4×FTE (demp_w4)	16.582 (6.557)	351	15.23 (6.23)
5	Weight part-time as 0.6×FTE (demp_w6)	16.144 (6.613)	351	14.60 (6.26)
6/8	Subsample regression (demp, N=289) — corresponds to either "exclude NJ shore area" (16.88) or "exclude stores called >2× in wave 1" (14.08)	15.204 (7.452)	289	16.88 / 14.08
10	Stores in towns around Newark (demp, N=351)	31.909 (10.095)	351	33.75 (16.75)

	Specification	Replicated gap (robust SE)	Rep. N	Original gap (SE)
12	Pennsylvania stores only / placebo (fake_gap)	−3.241 (22.394)	66	−0.30 (22.00)

Replication uses robust (heteroskedasticity-consistent) standard errors; the original uses conventional standard errors. This partly explains the larger standard errors in the replication.

The replication confirms the robustness of the positive wage-gap effect on employment across alternative FTE constructions. Coefficient estimates range from 15.79 to 16.58, marginally above the original benchmark of 14.92, and remain statistically significant. The placebo test on Pennsylvania stores with fake_gap yields an insignificant coefficient (−3.24, SE 22.39), closely matching the original (−0.30, SE 22.00), supporting the causal interpretation. Minor discrepancies stem from sample-selection differences and the use of robust standard errors, but qualitative conclusions are fully preserved.

4.5 Full-Time and Part-Time Substitution

TABLE 6—EFFECTS OF MINIMUM-WAGE INCREASE ON OTHER OUTCOMES

Outcome measure	Mean change in outcome: NJ (i)	PA (ii)	NJ − PA (iii)	Regression of change in outcome variable on: NJ dummy (iv)	Wage gap[a] (v)	Wage gap[b] (vi)
Store Characteristics:						
1. Fraction full-time workers[c] (percentage)	2.64 (1.71)	−4.65 (3.80)	7.29 (4.17)	7.30 (3.96)	33.64 (20.95)	20.28 (24.34)
2. Number of hours open per weekday	−0.00 (0.06)	0.11 (0.08)	−0.11 (0.10)	−0.11 (0.12)	−0.24 (0.65)	0.04 (0.76)
3. Number of cash registers	−0.04 (0.04)	0.13 (0.10)	−0.17 (0.11)	−0.18 (0.10)	−0.31 (0.53)	0.29 (0.62)
4. Number of cash registers open at 11:00 A.M.	−0.03 (0.05)	−0.20 (0.08)	0.17 (0.10)	0.17 (0.12)	0.15 (0.62)	−0.47 (0.74)
Employee Meal Programs:						
5. Low-price meal program (percentage)	−4.67 (2.65)	−1.28 (3.86)	−3.39 (4.68)	−2.01 (5.63)	−30.31 (29.80)	−33.15 (35.04)
6. Free meal program (percentage)	8.41 (2.17)	6.41 (3.33)	2.00 (3.97)	0.49 (4.50)	29.90 (23.75)	36.91 (27.90)
7. Combination of low-price and free meals (percentage)	−4.04 (1.98)	−5.13 (3.11)	1.09 (3.69)	1.20 (4.32)	−11.87 (22.87)	−19.19 (26.81)
Wage Profile:						
8. Time to first raise (weeks)	3.77 (0.89)	1.26 (1.97)	2.51 (2.16)	2.21 (2.03)	4.02 (10.81)	−5.10 (12.74)
9. Usual amount of first raise (cents)	−0.01 (0.01)	−0.02 (0.02)	0.01 (0.02)	0.01 (0.02)	0.03 (0.11)	0.03 (0.11)
10. Slope of wage profile (percent per week)	−0.10 (0.04)	−0.11 (0.09)	0.01 (0.10)	0.01 (0.10)	−0.09 (0.56)	−0.08 (0.57)

Variable	ΔFrac-Full (d_fracfull)	ΔHours-Open (d_hrsopen)	ΔReg-11am (d_nregs11)	ΔMeal-Prog (d_meals)	ΔTime-Raise (d_inctime)	ΔSize-Raise (d_firstinc)	ΔWage-Slope (d_slope)
gap	0.1841	0.0362	−0.4439	−0.8698*	−5.0872	0.0285	0.0149

Variable	ΔFrac-Full (d_fracfull)	ΔHours-Open (d_hrsopen)	ΔReg-11am (d_nregs11)	ΔMeal-Prog (d_meals)	ΔTime-Raise (d_inctime)	ΔSize-Raise (d_firstinc)	ΔWage-Slope (d_slope)
	(0.2419)	(0.7115)	(0.7878)	(0.4848)	(12.5883)	(0.1374)	(0.0227)
All controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	385	399	372	399	321	296	277

Except for employee-meal benefits, the gap coefficient is statistically insignificant at the 5 % level. Each regression has different sample sizes due to item-level missing values in survey responses for fringe-benefit and wage-policy questions.

Outcome variable	Original gap coefficient (SE)	Replication gap coefficient (robust SE)	Original significance	Replication significance	Replication N
Change in fraction full-time workers	33.64 (20.95)	0.1841 (0.2419)	Insig.	Insig.	385
Change in hours open per weekday	−0.24 (0.65)	0.0362 (0.7115)	Insig.	Insig.	399
Change in registers open at 11 a.m.	0.15 (0.62)	−0.4439 (0.7878)	Insig.	Insig.	372
Change in employee-meal program (composite)	−11.87 (22.87)	−0.8698* (0.4848)	Insig.	* p<0.10	399
Change in time to first raise (weeks)	4.02 (10.81)	−5.0872 (12.5883)	Insig.	Insig.	321
Change in size of first raise (cents)	0.03 (0.11)	0.0285 (0.1374)	Insig.	Insig.	296
Change in slope of wage profile	−0.09 (0.56)	0.0149 (0.0227)	Insig.	Insig.	277

Table compares the original estimates from Card and Krueger (1994) Table 6 against this replication. All regressions test whether restaurants facing a larger required wage increase (`gap`) adjusted non-employment margins such as worker composition, operating hours, service capacity, fringe benefits, or internal wage-setting rules.

Qualitatively, the replication largely supports the original conclusion: there is no statistically significant evidence at the 5 % level that minimum-wage-exposed restaurants altered most auxiliary outcomes. Coefficients are generally statistically insignificant in both the original and replicated models.

Minor discrepancies exist. First, coefficient magnitudes for the fraction-full-time variable differ substantially because the original uses percentage-point units while the replication uses decimal fractions. Second, signs differ for registers-at-11 a.m. and time-to-first-raise, though both original and

replicated coefficients are statistically insignificant, so these sign reversals carry little economic weight. Third, the composite employee-meal outcome is marginally significant at the 10 % level in the replication but insignificant in the original paper. This provides only weak suggestive evidence that firms may cut fringe benefits to offset higher wage costs.

Sample sizes vary across specifications in the replication, driven by substantial item-level non-response for fringe-benefit and wage-policy survey questions, consistent with the data limitations faced in the original study.

4.6 Price Effects of the Minimum-Wage Increase

TABLE 7—REDUCED-FORM MODELS FOR CHANGE IN THE PRICE OF A FULL MEAL

	Dependent variable: change in the log price of a full meal				
Independent variable	(i)	(ii)	(iii)	(iv)	(v)
1. New Jersey dummy	0.033 (0.014)	0.037 (0.014)	—	—	—
2. Initial wage gap[a]	—	—	0.077 (0.075)	0.146 (0.074)	0.063 (0.089)
3. Controls for chain and ownership[b]	no	yes	no	yes	yes
4. Controls for region[c]	no	no	no	no	yes
5. Standard error of regression	0.101	0.097	0.102	0.098	0.097

Specification	Regressors	gap coefficient (robust SE)	nj coefficient (robust SE)	N	R^2
(1) NJ dummy only	nj	—	0.0289** (0.0130)	356	0.0122
(2) NJ + chain & ownership controls	nj bk kfc roys co_owned	—	0.0316** (0.0130)	356	0.0625
(3) Gap only	gap	0.0858 (0.0627)	—	356	0.0041
(4) Gap + chain & ownership controls	gap bk kfc roys co_owned	0.1466*** (0.0544)	—	356	0.0594
(5) Full: Gap + chain + ownership + region dummies	gap bk kfc roys co_owned southj centralj northj pa1	0.0835 (0.0705)	—	356	0.0742

This table presents regression estimates for the log change in meal prices following New Jersey's minimum wage increase. The difference-in-differences specification with the New Jersey treatment dummy shows a statistically significant price increase of approximately 3% in New Jersey relative to Pennsylvania. The wage-gap model, which exploits cross-store variation in required wage rises, yields

a significant positive price effect after controlling for chain and ownership characteristics. However, this gap effect attenuates and becomes statistically insignificant once regional fixed effects are included. Taken together, the results suggest that the evidence for price pass-through is sensitive to regional controls, and the estimated price response is not robust to location adjustment.

Specification	Key coefficient	Original Coefficient (SE)	Replication Coefficient (robust SE)	N	Original R	Replication R
(i) NJ dummy, no controls	nj	0.033 (0.014)	0.029** (0.013)	315	0.010	0.012
(ii) NJ dummy, chain+ownership	nj	0.037** (0.014)	0.032** (0.013)	315	0.057	0.063
(iii) gap only, no controls	gap	0.077 (0.075)	0.086 (0.063)	315	0.002	0.004
(iv) gap, chain+ownership	gap	0.146** (0.074)	0.147*** (0.054)	315	0.054	0.059
(v) gap, chain+ownership+region	gap	0.063 (0.089)	0.084 (0.071)	315	0.073	0.074

The table reports regression results for minimum-wage pass-through to fast-food meal prices, replicating Table 7 from Card and Krueger (1994). Two identification strategies are adopted: a cross-state difference-in-differences using the New-Jersey dummy, and a within-New-Jersey wage-gap approach exploiting store-level variation in required wage increases.

The coefficient on the New-Jersey dummy is positive and statistically significant (i-ii). After controlling for chain and ownership characteristics, meal prices in New Jersey rose by approximately 3.2 % relative to Pennsylvania. This provides evidence of partial cost pass-through to consumers and is consistent with the original study.

Without controls (iii-v), the gap coefficient is positive but insignificant. After adding chain and ownership controls, gap becomes significantly positive. A 0.10 increase in wage gap (a required 10 % wage rise) corresponds to a 1.47 % faster growth in meal prices. However, once regional fixed-effects are included, the coefficient shrinks and loses statistical significance. Local geographic price variation absorbs most of the store-level price-pressure signal.

The evidence for price pass-through is mixed and sensitive to model specification. Cross-state comparison detects meaningful price growth in New Jersey. Nevertheless, the within-state evidence that more-exposed stores raise prices faster disappears after accounting for regional heterogeneity. This replicates the original authors' key conclusion.

4.7 Broader Evidence on Employment Changers in New Jersey

TABLE 8—ESTIMATED EFFECT OF MINIMUM WAGES ON NUMBERS OF MCDONALD'S RESTAURANTS, 1986–1991

Independent variable	Dependent variable: proportional increase in number of stores				Dependent variable: (number of newly opened stores) ÷ (number in 1986)			
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Minimum-Wage Variable:								
1. Fraction of retail workers in affected wage range 1986[a]	0.33 (0.20)	—	0.13 (0.19)	—	0.37 (0.22)	—	0.16 (0.21)	—
2. (State minimum wage in 1991) ÷ (average retail wage in 1986)[b]	—	0.38 (0.22)	—	0.47 (0.22)	—	0.47 (0.23)	—	0.56 (0.24)
Other Control Variables:								
3. Proportional growth in population, 1986–1991	—	—	0.88 (0.23)	1.03 (0.23)	—	—	0.86 (0.25)	1.04 (0.25)
4. Change in unemployment rates, 1986–1991	—	—	−1.78 (0.62)	−1.40 (0.61)	—	—	−1.85 (0.68)	−1.40 (0.65)
5. Standard error of regression	0.083	0.083	0.071	0.068	0.088	0.088	0.077	0.073

Table 8 of Card and Krueger (1994) relies on state-level data covering all US states from 1986 to 1991, which records the number of McDonald's restaurants, population and unemployment statistics. The dataset used in this replication (njmin.dta) is a restaurant-level micro survey limited to fast-food establishments in New Jersey and Pennsylvania only. It does not contain state-level aggregate information required to construct variables such as store_growth, frac_affected and mw_ratio. For this reason, Table 8 cannot be replicated and is excluded from the analysis.

5. Independent Extension：Heterogeneous Effects by Store Ownership

5.1 Results

In the original analysis of Card and Krueger (1994), the store-ownership dummy co_owned is only included as one of the control variables, without explicitly testing heterogeneous responses between company-owned and franchise stores.

Franchise outlets are operated by private local owners who bear residual profit risk and may be more sensitive to rising labour costs under the competitive labour-market framework. By contrast, company-owned stores are managed by chain headquarters and face relatively soft budget constraints. If the competitive model holds, franchise stores would exhibit stronger employment reduction or larger price increases when hit by a minimum-wage shock. Under monopsony logic, however, the difference across ownership types would be muted.

This extension constructs the interaction term gap × co_owned and implements both pooled interactive regressions and split-sample subgroup regressions. Two outcome variables are studied: change in full-time-equivalent employment (d_fte) as in Table 4, and log-change in full-meal price (dln_pmeal) as in Table 7. The same set of chain dummies, regional dummies and ownership indicator are maintained to be consistent with baseline specifications.

Variable	Pooled ΔFTE	Company-owned ΔFTE	Franchise ΔFTE	Pooled Δln(meal price)
gap	11.462 (8.780)	-3.282 (9.596)	13.066 (9.403)	-0.006 (0.085)
co_owned	1.355 (1.393)	—	—	-0.021 (0.019)
gap × co_owned	-14.059 (11.481)	—	—	0.288** (0.134)
Chain & region controls	Yes	Yes	Yes	Yes
Observations	385	134	251	356

For employment changes (d_fte = fte2-fte1), the pooled regression returns a positive but insignificant gap coefficient. The interaction term gap_coowned is negative and statistically insignificant. Subgroup regressions show a positive, marginally insignificant gap coefficient for franchise stores and a small negative, insignificant coefficient for company-owned stores. There is no statistically significant evidence that ownership structure generates different employment responses to the minimum wage rise.

For meal price growth (dln_pmeal), the interaction term gap_coowned is positive and significant at the 5% level. The total marginal effect of gap for company-owned stores is approximately 0.282, while the effect for franchise stores is nearly zero. This result indicates that price increases in response to minimum wage pressure are concentrated exclusively in company-owned restaurants. This pattern is inconsistent with simple competitive predictions, under which franchise owners with tighter profit margins would be more likely to raise prices. A plausible interpretation is that chain headquarters stabilise employment at their own stores but pass higher labour costs to consumers through menu price adjustments, whereas franchise stores may rely on alternative adjustment channels outside of menu prices.

5.2 Validation

Two empirical strategies — pooled regression with interaction term and separate subgroup regressions — are used to cross-validate the heterogeneity results. All model specifications keep the exact control set from the baseline replication of Table 4 and Table 7. The estimation outcomes are compared against the original gap-based baseline estimates to judge whether ownership structure modifies the minimum-wage treatment effect. Note that Table 8 cannot be examined in this extension because of missing external state-level data.

5.3 Limitations

First, the subsample size of company-owned stores is 134, which reduces statistical power to detect small heterogeneous employment effects. Second, ownership status may correlate with other unobserved store characteristics such as

local demand conditions, so observed price differences cannot be purely attributed to ownership incentive alone. Third, this heterogeneity analysis is restricted to fast-food chain stores in New Jersey and eastern Pennsylvania; findings cannot be generalised to independent small restaurants, other industries or other geographic contexts. Finally, the price finding is only from the pooled interaction regression; separate subgroup price regressions would further test robustness.

6. Conclusion

This paper replicates the core establishment-level analysis of Card and Krueger (1994). Consistent with the original work, baseline difference-in-differences and wage-gap regressions show no evidence that New Jersey's minimum-wage hike reduced full-time-equivalent employment. Cross-state comparisons detect moderate price increases in New Jersey, though within-state price pass-through is not robust to regional fixed-effects. Non-wage benefit regressions are largely insignificant, offering little support for cost offset via fringe-benefit cuts.

An independent extension investigates heterogeneity between company-owned and franchise stores. Employment responses do not statistically differ across ownership types, consistent with monopsony-type expectations. However, price adjustment differs: only company-owned stores pass higher labor costs to consumers by raising meal prices, while franchise stores leave menu prices nearly unchanged.

Important caveats remain. The sample is restricted to fast-food chains in two US states, limiting external validity. Survey measurement error may bias estimates toward zero. Due to data constraints, Table 8 on store openings cannot be replicated, so we cannot assess effects on firm entry. While the results confirm that moderate minimum-wage increases do not necessarily hurt employment for existing restaurants, firms' cost-adjustment strategies vary with organizational form, and findings should not be generalized carelessly to broader labor markets.

7. AI-Use Disclosure

7.1 Where and how AI was used

AI was used to draft Stata code for the independent heterogeneity extension across store ownership types, generate regression table templates, structure the empirical interpretation of estimation outputs, and refine academic writing for the extension section and conclusion. AI helped organise the logical framework for the replication report and suggested ways to align the discussion with Card and Krueger's original theoretical arguments. All raw regression estimation was run independently by me in Stata using the dataset.

7.2 How AI-generated code, analysis, and writing were checked

All AI-provided Stata syntax was manually tested in Stata. I cross-checked the generated regression outputs against the raw Stata results, verified coefficient values, standard errors and p-values. AI-drafted text was reviewed line by line to ensure consistency with my empirical findings. I also compared the analysis against the original Card and Krueger (1994) paper to confirm the economic interpretation

matched the research context. All final numbers and claims in the report are based on my own Stata runs.

7.3 Important AI errors or suggestion that you corrected or rejected

(1) AI initially miscalculated the DID first (2.75 but right is 2.69); I manually recalculated and corrected this value.

(2) AI's first draft omitted the requirement to replicate Table 6 (non-wage benefits) from the original paper. I rejected this omission and added Table 6 replication and discussion into the report structure.

(3) Some early AI interpretations overstated the causal implication of the significant price interaction term; I revised the wording to emphasize that the result is correlational and limited by the dataset's scope.